## READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

# A Short History of Glass

**A** Long before people learned to make it, they used glass that nature had made. Obsidian, a black glass that forms when lava cools very quickly, was chipped into knives and arrowheads more than ten thousand years ago. Because it breaks into pieces with extremely sharp edges, it was valued so highly that it was carried and traded hundreds of kilometres from the volcanoes where it was found.

**B** The first glass made by people appeared in Mesopotamia around 3500 BC, probably as a by-product of metalworking or pottery. For many centuries it was used only for small objects such as beads and the shiny coating on clay pots. Making a vessel was slow and costly: workers wrapped hot glass around a core of sand and clay, and then scraped the core out once the glass had cooled.

**C** Everything changed in the first century BC, when craftsmen on the eastern coast of the Mediterranean discovered that glass could be blown. By blowing air through a metal pipe into a small ball of molten glass, a worker could form a thin bottle in minutes. Glass became cheap enough for ordinary households, and the Romans spread the technique to every corner of their empire.

**D** In medieval Europe, glassmakers were both admired and controlled. In Venice the industry was moved to the island of Murano in 1291, officially to protect the city from fire. The move also kept the secrets of the trade in one place: skilled workers were treated as important citizens, but they were not allowed to leave the republic.

**E** Clear window glass remained a luxury until the nineteenth century. Early panes were made by spinning a disc of glass, which left them thick in the middle and uneven at the edges. In 1959, the British engineer Alastair Pilkington introduced the float process, in which molten glass is poured onto a bath of liquid tin. The result is a perfectly flat sheet, and almost all window glass is still made this way today.

**F** Today glass is found in places its first makers could never have imagined. Thin strands of glass fibre carry most of the world’s internet traffic as pulses of light, and scientists are developing glass that can store data for thousands of years. Yet the basic recipe has hardly changed: sand, soda and lime, heated until they melt.

## Questions 1–4

Do the following statements agree with the information given in Reading Passage 1?
In boxes 1–4 on your answer sheet, write

| | |
|---|---|
| TRUE | if the statement agrees with the information |
| FALSE | if the statement contradicts the information |
| NOT GIVEN | if there is no information on this |

**1** Obsidian was traded far from the places where it was found.

**2** The first man-made glass was produced deliberately by potters.

**3** Glass blowing made glass objects affordable for ordinary families.

**4** Venetian glassworkers were paid more than other craftsmen.

## Questions 5–8

Reading Passage 1 has six paragraphs, A–F.
Which paragraph contains the following information?
Write the correct letter, A–F, in boxes 5–8 on your answer sheet.

**5** a modern use of glass that the earliest makers would not have expected

**6** a reason given for moving an industry to a new place

**7** how vessels were made before a new technique was found

**8** a method that produces completely flat glass

## Questions 9–10

Choose the correct letter, A, B, C or D.

**9** Why was obsidian valued?

- **A** it was easy to find near volcanoes
- **B** it could be melted and reshaped
- **C** it produced very sharp edges
- **D** it was the only glass that was black

**10** What does the writer say about the float process?

- **A** it was invented in Venice
- **B** it is still the usual way of making windows
- **C** it made glass thicker in the middle
- **D** it replaced glass fibre

## Questions 11–12

Choose TWO letters, A–E.
Which TWO statements about the glass industry in Venice are true?

- **A** It started in the first century BC.
- **B** It was moved to an island.
- **C** Its workers could not leave the republic.
- **D** It sold glass to the Romans.
- **E** It invented glass blowing.

## Question 13

Complete the sentence below.
Choose ONE WORD ONLY from the passage for your answer.

The basic recipe for glass is sand, soda and 13 ……

## ANSWERS

1 TRUE — paragraph A: “traded hundreds of kilometres from the volcanoes”

2 FALSE — paragraph B: “probably as a by-product of metalworking or pottery”

3 TRUE — paragraph C: “cheap enough for ordinary households”

4 NOT GIVEN — the passage does not compare their pay

5 F — glass fibre carries internet traffic

6 D — “officially to protect the city from fire”

7 B — a core of sand and clay was used

8 E — “a perfectly flat sheet”

9 C — “extremely sharp edges”

10 B — “almost all window glass is still made this way today”

11–12 IN EITHER ORDER B C — paragraph D

13 lime — paragraph F